ART. IV.—*Double-Star Discoveries with the* 18½*-inch Chicago Refractor;* by S. W. BURNHAM.

THE new double stars described below were discovered with the 18½-inch Clark refractor of the Dearborn Observatory, on the nights of October 1, 2, 4, 7, 10, 11, 15, 16, and 17, 1876, and are some of the results obtained in the use of that instrument during a period of three or four weeks, the only opportunity the writer has had of using it. The observations here given are not as complete as could be desired; some of the new pairs being only partially measured, and others not at all. A number of other new and interesting pairs were found, but the approximate readings of the circles, roughly noted at the time of finding them, are not sufficient to positively identify them in the star catalogues and thus fix their absolute places; and these with other suspected pairs, and new companions to prominent pairs and binary systems, are necessarily not included in the following list. The intention was to measure, during the course of the work, all new pairs on at least three different nights, in order to give, as accurately as possible, reliable micrometrical results with which future observations might be compared for determining the question of physical relation between members of the systems. This seemed to be the more important from the fact that many of these pairs are too difficult for ordinary apertures, and are practically beyond the reach of most of the well-known double-star observers in Europe; and a considerable time might elapse before the objects would be re-observed elsewhere. It is necessary to state, in explanation of its manifest incompleteness, that the work was unexpectedly, and without notice, terminated by the action of the officers of the Chicago Astronomical Society which has control of the instrument, and no opportunity has been afforded to complete the projected series of observations which was but just commenced. Enough is given, however, to show the value and effectiveness of the telescope in this important branch of astronomical research; and how much might be accomplished for science, if it were used for any other purpose than exhibition to visitors.

The reference numbers attached to these stars are the numbers in my double-star catalogues, seven of which lists, with the stars numbered consecutively, have appeared in the Monthly Notices of the Royal Astronomical Society, Astronomische Nachrichten, and in this Journal.

No. 437=L 4291.

$$\left.\begin{array}{l}\text{R. A.}=2^h\ 12^m\ 26^s\\ \text{Decl.}=+3^\circ\ 39'\end{array}\right\}$$

Not measured, but angle and distance estimated as follows:

P=45°± D=5″±

The principal star is about 7½ magnitude, and the companion not brighter than 11 of Struve's scale. The pair of small stars in the field, *sp* is Σ 247 *rej.*

No. 438=Σ 2538

R. A.=19ʰ 27ᵐ 3ˢ
Decl.=+36°27′

A and B	P=43°·5	D=6″·32	(1876·8)
A and C	245·2	53·04	(1830·8)
C and D	52·5	6·07	(1830·8)

The three large stars, A, C and D, constitute the double star Σ 2538(=S 719), but the small companion near the principal star is now recorded for the first time, and is probably too minute for ordinary telescopes. The measures given above of C and D are by Struve. These stars seem, by a comparison of Struve's results with those obtained later by Mädler, Secchi and Otto Struve, to be relatively fixed, but some of the measures are not very accordant. By a single observation I found these angles respectively, 237°·1 and 51°·9, but there may be an error of 10° in reading micrometer of the first. There is still another new member of this group, in an exceedingly faint star almost exactly midway between A and C. No opportunity occurred to measure this. Struve gives the magnitudes of A, C and D as 8·2, 8·3 and 8·7 respectively.

No. 439=Arg. (29°) 3845

R. A.=19ʰ 55ᵐ 56ˢ
Decl.=+29° 30′

The principal star is about the eighth magnitude (Argelander, 8·1), with a small companion. The measures of one night give:

P=249°·7 D=2″·70 (1876·8)

No. 440=L 38520

R. A=20ʰ 1ᵐ 27ˢ
Decl.=+35° 27′

A and B	P=61°·3	D=6″·47	Mags. 7·0 . . 12+	(1876·8)
A and C	25·8	7·75	11·0	(1876·7)
A and D	296·0	11·27	9·5	(1783·7)
A and E	106·8	28·15	11·5	(1876·7)
A and F	32·8	29·45	7·5	(1783·7)
F and G	113·0	10·12	12·	(1876·8)

The large stars (A, D, F), of this interesting group have long been known, and constitute the double star ♃ III. 113 (=Sh 314=Σ 2630 *rej*). The small attendants, C and E, I